Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/21/2006
PI Name:	Butler, Bruce D. Ph.D.		
Project Title:	Effects of Circadian Rhythms on Decompression Outcome		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Environmental health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	07/01/1998	**End Date:**	03/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smolensky, Michael (The University of Texas at Austin)		
Grant/Contract No.:	NAG 9-1040		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Astronauts undertaking extravehicular activities were subjected to decompression. In this project we examined the effects of the body's endogenous circadian rhythms on the physiological consequences of decompression-induced venous bubble formation. Specific indices of inflammation and of lung injury were previously reported with decompression sickness, even with mild to moderate levels of decompression. Based on these observations, we studied the effect of the circadian rhythms on the tolerance to decompression exposure and the underlying 24 hour bioperiodic dependencies. Our overall objectives included the quantitative evaluation of the physiological stress of decompression at different circadian times and to relate these to rhythms in cellular and biochemical variations over the 24 hr. span. Measurement of these responses may elucidate some of the less understood risks associated with decompression illness, and may lead directly to operationally useful procedures.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The role of circadian rhythms in human performance and disease processes is of great importance to many applications of commerce and health. Such factors are considered with operational requirements for factory shift workers, airline pilots, and many other civilian careers as well as in the military where maneuvers can occur throughout a 24 hour day. The application of circadian rhythm influences to decompression stress has not been thoroughly evaluated in the context of NASA astronauts or with regards to other activities where decompression is a concern such as sport, commercial and military scuba divers or aviation workers apart from astronauts. Benefits derived from this research will potentially identify the principal times where Decompression sickness (DCS) risk may be influenced by circadian rhythms and therefore allow decisions to be made regarding appropriate scheduling of decompression exposures. Additional benefits may derive from the pharmacological studies proposed that will elucidate the potential dosing effects that may vary with time of day administration.
Task Progress:	Decompression-induced venous bubbles can lead to obstruction of blood vessels, alteration in vessel wall permeability and vasoconstriction. Apart from the mechanical obstruction induced by the bubbles, these effects are partially caused by release of arachidonic acid metabolites, such as thromboxanes and leukotrienes, resulting from bubble-activated neutrophils and endothelial cell injury. Circadian rhythm-dependencies in the tolerance to DCS were evident in most of the parameters studied. These observations concur with other studies examining the circadian rhythm-dependencies of inflammatory-like reactions to various stresses. Pharmacological intervention on the effects of inflammatory-like responses to DCS showed reductions, however, the hemodynamic and mechanical effects of the intravascular bubbles may also require specific intervention. It is concluded that DCS risk is affected by the circadian time structure and should be taken into consideration for properly assessing the underlying mechanisms of DCS as well as the efficiency of preventive and interventions strategies.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Little T, Butler BD. "Pharmacological intervention to the inflammatory response from decompression sickness in rats." Aviat Space Environ Med. 2008 Feb;79(2):87-93. PubMed PMID: 18309904 , Feb-2008